

$E_3 A = \begin{bmatrix} a & b & c \\ d & e & f \\ 5g & 5h & 5i \end{bmatrix}$. Addition of -4 times row 1 of A to row 3 produces $E_4 A$. (This is a row replacement operation.)

An interchange of rows 1 and 2 of A produces $E_2 A$, and multiplication of row 3 of A by 5 produces $E_3 A$. $\square$

Left-multiplication (that is, multiplication on the left) by E_i in Example 5 has the same effect on any $3 \times n$ matrix. It adds -4 times row 1 to row 3. In particular, since $E_i \cdot I = E_i$, we see that E_i itself is produced by this same row operation on the identity.

General fact

If an elementary row operation is performed on an $m \times n$ matrix A , the resulting matrix can be written as EA , where the $m \times m$ matrix E is created by performing the same row operation on I_m .

The proof will be done in Exercises 27 and 28.

Since row operations are reversible, as shown in Section 1.1, elementary matrices are invertible, for if E is produced by a row operation on I , then there is another row operation of the same type that changes E back into I . Hence there is an elementary matrix F such that $FE = I$. Since E and F correspond to reverse operations, $EF = I$, too.

Each elementary matrix E is invertible. The inverse of E is the elementary matrix of the same type that transforms E back into I .

Example 6: Find the inverse of $E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$

Solution: To transform E_1 into I , add $+4$ times row 1 to row 3. The elementary matrix that does this is $E_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$ $\square$

Theorem 7: An $n \times n$ matrix is invertible if and only if A is row equivalent to I_n , and in this case, any sequence of elementary row operations that reduces A to I_n also transforms I_n into A^{-1} .